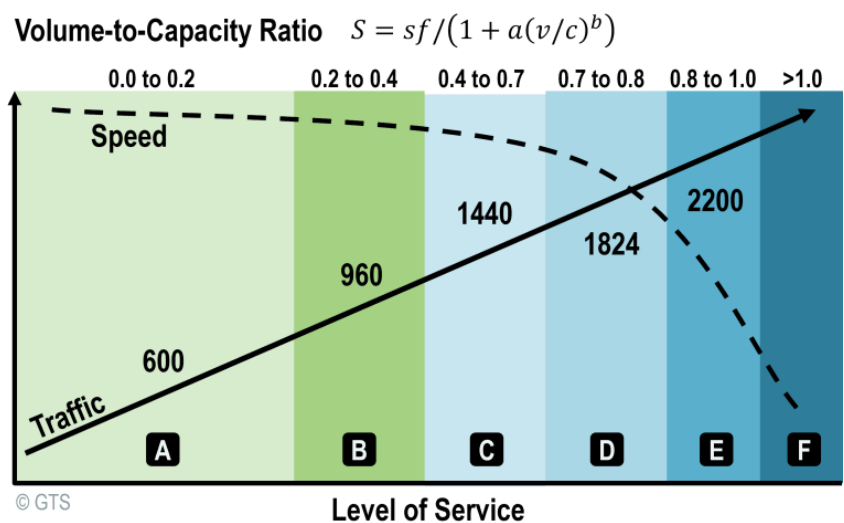




Levels of Service for Road Transportation

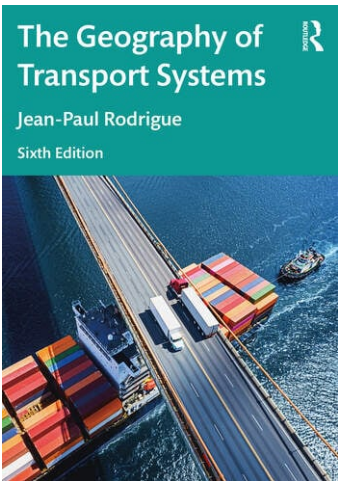


Levels of Service for Road Transportation

Source: *Transportation Research Board (1994) Highway Capacity Manual, 3rd Edition*. sf = free flow speed, v = volume, c = capacity, $a = 0.15$ and $b = 4$.

There is a relationship between traffic speed, volume and density for a highway, and how these factors relate to Level of Service ratings. Traffic speed and flow on urban streets are determined primarily by intersection capacity, which is affected by traffic volumes on cross streets and left turn signal phases. As these tables indicate, traffic congestion is a non-linear function, meaning that a small reduction in urban-peak traffic

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[The Geography of Transport Systems](#)
SIXTH EDITION
Jean-Paul Rodrigue
(2024), New York:
Routledge, 402 pages.
ISBN 9781032380407

DOI:
[10.4324/97810033431](https://doi.org/10.4324/97810033431)

volume can cause a proportionally larger reduction in delay. For example, a 5% reduction in traffic volumes on a congested highway (for example, from 2,000 to 1,900 vehicles per hour) may cause a 10-30% increase in average vehicle speeds (for example, increasing traffic speeds from 35 to 45 miles per hour). As a result, even relatively small changes in traffic volume or capacity on congested roads can provide relatively large reductions in traffic delays.

LOS	Speed Range (mph)	Flow Range (veh./hour/lane)	Density Range (veh./mile)
A	Over 60	Under 700	Under 12
B	57-60	700-1,100	20-Dec
C	54-57	1,100-1,550	20-30
D	46-54	1,550-1,850	30-42
E	30-46	1,850-2,000	42-67
F	Under 30	Unstable	67

LOS	A	B	C	D	E
4-lane Freeway	700	1,100	1,550	1,850	2,000
2-lane Highway	210	375	600	900	1,400
4-lane Highway	720	1,200	1,650	1,940	2,200

- **(A) Free Flow Traffic.** Users are practically unaffected by the presence of other vehicles on a road section. The choice of speed and the

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maneuverability are free. The level of comfort is excellent, as drivers need minimal attention. The volume-to-capacity ratio is usually below 0.2.

- **(B) Steady Traffic.** The presence of other vehicles begins to affect the behavior of individual drivers. The choice of the speed is free, but the maneuverability has somewhat decreased. The comfort is excellent, as drivers simply need to keep an eye on nearby vehicles.
- **(C) Steady Traffic but Limited.** The presence of other vehicles affects drivers. The choice of the speed is affected and maneuvering requires vigilance. The level of comfort decreases quickly at this level, because drivers have a growing impression of being caught between other vehicles.
- **(D) Steady Traffic at High Density.** The speed and the maneuverability are severely reduced. Low level of comfort for drivers, as collisions with other vehicles, must constantly be avoided. A slight increase in the traffic risks causing some operational problems and saturating the network.
- **(E) Traffic at Saturation.** Low but uniform speed. Maneuverability is possible only under constraint for another vehicle. Users are in a state of frustration.
- **(F) Congestion.** Unstable speed with the formation of waiting lines at several points. Cycles of stop and departure with no apparent pattern because created by the behavior of other drivers. A high level of vigilance is required for the user with practically no comfort. At this level, the volume-to-capacity ratio exceeds 1, implying that the road segment is used above design capacity.

The **rate of traffic service** is the maximal hourly rate that can cross a point or a road section according to road, traffic, and control conditions. Therefore, each road infrastructure has five traffic rates of service (level

F is not used because unstable). Traffic reports also use color codes to illustrate traffic conditions, such as green (levels A and B), yellow (levels C and D), and red (levels E and F).

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